CALIBRATION LABORATORY GUIDE

Standard Metrology Edition

Pilot / Early Access

The Asset operating foundation developed for calibration laboratories: customer work, technical evidence, review and accountable service.

Equipment. People. The work around them.

Asset and AssetLight provide the general foundation. Standard develops that foundation for industry-specific work; Metrology is the principal current edition.

Agree product configuration, evaluation scenarios and adoption terms before rollout.

PRODUCT FIT

Choose the scope around your work.

The products share a foundation. Specialization and the agreed operating scope determine the fit.

Asset / Pilot / Early Access	General-purpose instrumentation and business management. Includes the AssetLight foundation, with customer work, purchasing, people and invoicing in its intended general business scope.
AssetLight / Pilot / Early Access	Focused instrumentation management. Equipment identity, location, service history and supporting records. Confirm the import and employee workflows needed for your pilot.
Standard Metrology / Pilot / Early Access	Calibration laboratory workflows on the Asset foundation. CMC budgets, supported point uncertainty, calibration records, technical review, certificates and customer service.
Future editions	Standard Public Safety is in development. Standard Traceability is in design. Their publications describe intended direction.

Current scope

Agree the configuration, evaluation scenarios and adoption terms with MetraSlate before rollout. Package exposure and accepted customer workflows must be confirmed for the selected pilot.

FROM REQUEST TO RESULT

A completed measurement is not a completed job.

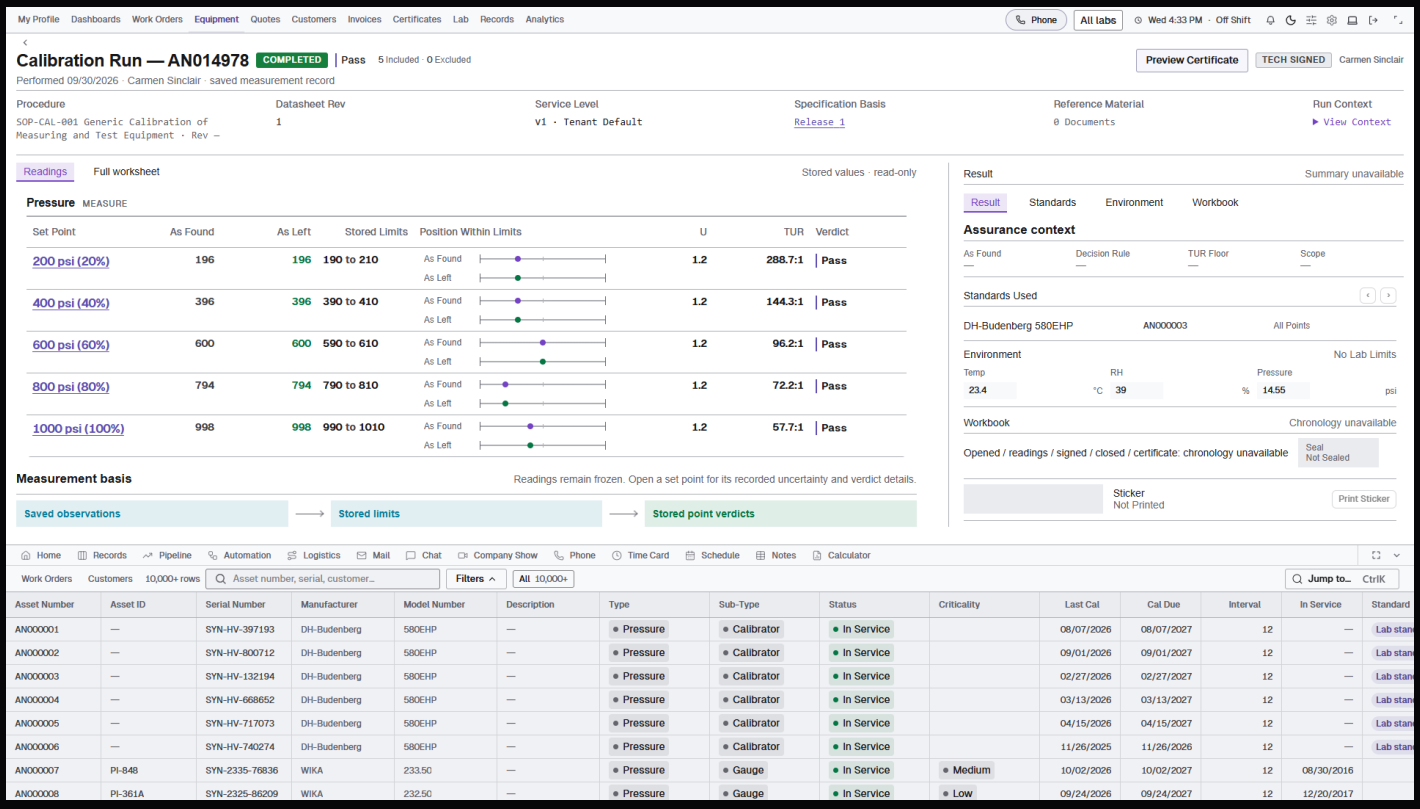
The customer agreement, instrument, technical result and released documents each carry a different responsibility.

Request and agree	Identify the customer equipment, service, estimate revision and customer decision. Keep PO and policy context with the work.
Receive and prepare	Connect intake and custody to instrument-level order lines. Identify the method, reference standards and required preparation.
Measure and evaluate	Record as-found and as-left observations. Inspect supported standard contributions, DUT resolution, coverage and decision policy.
Review and release	Keep technical completion, accountable review, certificate generation and customer release distinct. Verify the complete role workflow in the pilot.
Return and reconcile	Follow the handover, invoice, recorded receipts and account balance. Active payment processing is not part of the offer described here.

EXISTING DEMONSTRATION

Inspect the observation and its evaluation.

As-found and as-left measurements, limits and attributed standards belong to the calibration record.



A Standard Metrology measurement workspace provides the setting for reviewing the recorded values and their calculation basis.

Existing Standard Metrology demonstration capture. This image does not establish an accepted Asset configuration, live customer data or current deployment availability.

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and uncertainty budgets give a capability claim its numerical basis.

Create scope entries and contributor budgets, calculate supported endpoints, and publish a reusable technical curve. Keep numerical support separate from accreditation approval.

Pilot / Early Access. CMC budgets and technical curve publication are part of the Metrology pilot discussion. Broader scope lifecycle and accreditation treatment remain in validation.

Standard brings CMC budget calculation, scope entries and measurement uncertainty into the calibration workflow. A technical manager can maintain the evidence behind a claimed range while technicians and reviewers inspect the basis used for individual measurements.

The work begins with a defined reference standard and measurement function. The laboratory records the source or measure role, range, unit, optional frequency band, and claimed gain, offset and coverage factor. This gives the capability a numerical definition that can be checked against its supporting budget.

Workflow and evidence

The scope entry identifies what is claimed.	Reference-standard equipment, function, role, range, units and frequency qualification keep the claim attached to the capability it describes.
Each contributor retains its assumptions.	Low and high values, sensitivity, distribution, input coverage, degrees of freedom and exclusion state explain how the budget was assembled.
The endpoints support a reusable curve.	Standard calculates the endpoint budgets and fits a gain-plus-offset uncertainty curve for the declared band.
The claim can be checked against its support.	Claim states distinguish an uncomputed budget, a claim supported by its budget and a claim that is tighter than the configured evidence supports.

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

Publication creates a controlled specification release.	The published curve carries its budget reference and coverage factor. Republishing replaces that budget's previous curve within one release.
The laboratory keeps responsibility for accreditation.	Standard organizes and calculates the supporting evidence. The laboratory and its accrediting body approve the scope and accreditation.

The contributor budget explains the capability.

A CMC budget retains contributor values at a low and a high point. Sensitivity and the selected distribution convert those inputs into standard uncertainty contributions. Supported inputs include normal-expanded, rectangular, triangular, U-shaped and standard-deviation values. An expanded-uncertainty input retains the coverage factor at which it was stated.

Standard combines the uncorrelated contributions by root-sum-square and calculates effective degrees of freedom. The scope-budget calculation can use a fixed coverage factor or its Student-t coverage basis. It applies one coverage factor across the two endpoints and rounds the expanded endpoint uncertainties upward to the supported publication precision before fitting the curve.

The claim and its supporting calculation remain connected.

A capability claim can become unsupported when a contributor changes. Standard distinguishes a claim that has no computed budget from one supported by the calculation and one that undercuts its supporting uncertainty. A refusal identifies the numerical disagreement so the laboratory can review the claim or its contributors.

Publishing places the fitted curve into the reference standard’s specification catalog. The curve retains its coverage and budget relationship. This is a technical publication; it is not an accrediting body’s approval of the laboratory’s scope.

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

A built-in CMC calculator makes the arithmetic visible.

The engineering-calculator library also includes a CMC Uncertainty Budget tool. Its survey, low-point budget, high-point budget and equation views let the user examine contributor values, sensitivity, distributions, degrees of freedom, endpoint uncertainty and the fitted gain and offset.

Calculator drafts and saved results are stored locally in the user's browser. A calculator study is useful for examining the numbers; the scope-entry budget and controlled publication keep the laboratory's reusable capability definition attached to its reference standard.

Capability work continues into measurement review.

A published reference-standard curve supplies a contribution to later point evaluation. The calibration record then combines the actual attributed standards with the DUT-resolution contribution and the selected measurement policy. This connects capability preparation to the uncertainty reported for the work.

The laboratory can revisit contributors and ranges as its standards and methods change, while reviewing completed results against their retained calculation evidence. Budget authoring supports preparation and maintenance of scope evidence; it does not automatically submit an accrediting body's complete scope document.

Scope Review and Accreditation Boundaries

Scope submission, review and publication controls are in development and validation. The technical budget and the laboratory’s external accreditation have distinct purposes.

Effective-date resolution across completed results, full-range quantitative validation and result-level accredited treatment on mixed certificates require further validation. Confirm these requirements explicitly in the pilot; a computed endpoint or published curve does not establish them.

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/cmc

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty budgets keep the reported result explainable.

Follow the actual standard contributions, resolution, combination, coverage and decision basis from the recorded observation to the reported uncertainty.

Pilot / Early Access. Supported point uncertainty is demonstrated in Standard Metrology. Repeated-observation completion and general correlated-input propagation remain outside the validated scope described here.

Standard performs the uncertainty evaluation for its calibration record. It resolves the contributions appropriate to the measurement, adds the configured DUT-resolution contribution and calculates the result under the selected service policy. The technician and reviewer can inspect the calculation at the point where it is used.

The result retains more than a final uncertainty number. Contributions, units, sensitivity, specification identity, coverage, TUR, guardband, decision outcome and reporting values provide the evidence needed to understand how the laboratory reached its conclusion.

Workflow and evidence

The attributed standards determine the contribution basis.	Standard resolves the relevant specification and retains the curve, units, operating coordinate, distribution or divisor, sensitivity and resulting contribution.
Resolution is part of the point budget.	The configured DUT resolution adds its contribution through a rectangular, triangular or explicitly configured divisor treatment.
Independent Contributions and Coverage	Combine supported independent contributions with their sensitivities. Inspect coverage, decision policy and reporting precision in the point result.
The reported value retains its calculation basis.	Full calculated expanded uncertainty and the reported uncertainty remain distinct, so reporting precision does not replace the underlying result.

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

The decision uses the laboratory's configured policy.	Coverage, TUR basis, guardband, acceptance limits, gates and the optional CMC-floor check form part of the evaluation context.
Completion preserves the evidence used.	The finished record keeps observations, component values, specification snapshots and decision evidence together for review and certificate lineage.

A reviewer can trace each contribution.

A reference standard's contribution begins with its attributed identity and the specification applicable to the point. The retained trace shows the authored curve, the operating value and unit, coverage or distribution treatment, sensitivity and the contribution carried into the DUT quantity.

When the standard measures a different quantity, the declared measurement relationship supplies the transfer. A missing relationship or missing specification produces a named refusal. The evaluator does not present an incomplete contribution set as a complete uncertainty budget.

The arithmetic can be inspected at the measurement point.

The point inspector separates setup from calculation. Its calculation view shows reference-standard contributions, process uncertainty, the DUT-resolution contribution, combined standard uncertainty, coverage and reported expanded uncertainty. The same view places limits, guardband, TUR and the decision basis alongside the calculation.

For example, independent standard uncertainties of 1.20, 0.80 and 0.60 microvolts combine to approximately 1.56 microvolts. With fixed $k = 2$, the expanded uncertainty is approximately 3.12 microvolts before the selected reporting convention. The laboratory approves the contributors, sensitivity assumptions and coverage appropriate to its method.

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

The measurement and its decision retain their policy.

Service policy controls how the supported uncertainty model reports its result and how the decision uses it. Standard retains the coverage factor, coverage probability, TUR basis, guardband, acceptance interval and the resulting outcome rather than collapsing them into a single pass flag.

The optional CMC-floor check compares reported uncertainty with a matching working-scope claim. It is not a complete accreditation determination or proof of effective-date treatment throughout a certificate.

Completed calculations preserve the basis used at the time.

Completion retains the raw and evaluated readings, contribution rows, calculation values, policy results and selected specification snapshots with the event. Later reference-data changes therefore do not silently replace the evidence behind the completed result. Historical records can also identify calculation detail that was not recorded at the time.

External observations require an agreed and validated ingestion contract before they can participate in this workflow. A measurement integration must preserve source identity and units; no universal instrument connector is implied.

Repeated Readings and Correlated Inputs

Repeated-reading summaries and statistical reference material are distinct from completed calibration uncertainty. Observation-set calculation and completion, general covariance propagation and uncertainty of an estimated sample spread are not included as validated pilot capabilities.

If your method depends on repeated observations or shared effects, bring the method and acceptance cases to the technical evaluation. An ordinary spread statistic must not be presented as its uncertainty.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/uncertainty

MANUAL CALIBRATION

Standard captures and evaluates the technician's measurements.

Standard provides the manual-calibration worksheet, measurement evaluation and technical record for laboratory work.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

The technician starts from the individual instrument and its service line, then opens the applicable manual-calibration event. Standard supplies the approved measurement points and technical context. The technician enters observations with their units and records the reference standards, environmental conditions, adjustment and notes required for the work.

As-found and as-left observations remain distinct. Standard evaluates the points against the configured specifications and service policy, calculates measurement uncertainty and TUR, and presents the applicable measurement analysis and decision result. The technician can inspect the basis of the evaluation while reviewing the instrument's received and returned conditions.

Completion preserves the observations and the technical snapshot used to evaluate them. Technician signing, technical review, document generation and customer release remain accountable steps. The completed event connects the measurement evidence to the instrument, work order and certificate, so a reviewer can retrieve the basis of the result later.

Related website guide: metraslate.com/metrology/manual-calibration

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Evaluate the specification that applies to the instrument's operating conditions.

Connect temperature, elapsed time, warm-up, mode, frequency, loading and recorded environmental conditions to controlled specifications and instrument-specific corrections.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

An accuracy statement is meaningful within its stated operating conditions. Standard retains those conditions with the model, selects the applicable specification for the point and exposes the calculation used to turn that specification into a contribution to the measurement result.

Manufacturer limits, instance calibration corrections, environmental influences and DUT acceptance limits have different roles. Keeping their provenance, validity and uncertainty explicit helps the laboratory apply the right information without silently substituting one for another.

Workflow and evidence

Temperature terms keep their defined reference.	Retain the base accuracy, reference temperature or band, valid temperature limits, coefficient form and the recorded temperature used in the evaluation.
Operating conditions determine applicability.	Use time since calibration, warm-up, autocalibration, resolution, mode, frequency, loading and other stated conditions to resolve the specification that governs the point.
Environmental observations retain their context.	Record conditions by run or point with source, timestamp and units, and identify missing, changing or unsuitable observations.
Instance corrections retain their certificate evidence.	Keep instrument-specific coefficients and certificate-derived corrections separate from manufacturer limits, with validity, uncertainty, conditions and revision.

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Conditional specifications and instance corrections / continued

Piecewise boundaries resolve deterministically.	Make overlaps, gaps, endpoint inclusion and precedence explicit so the same inputs select the same applicable rule.
Draft recomputation preserves released history.	Changes invalidate dependent draft results and trigger recomputation, while released evidence continues through controlled amendment and lineage.

Temperature-dependent accuracy retains its formula and valid range.

A temperature model records the base accuracy, reference temperature or reference band, valid temperature limits and coefficient form. The author can express supported terms as ppm, a percentage of reading or range, an absolute quantity or an accuracy multiplier, with the applicable function and range identified.

The evaluation retains the observed temperature, its source and time, the coefficient, the resulting contribution and the specification revision. Calibration temperature and autocalibration temperature are separate conditions where the authored specification uses them. The model identifies how each term applies and avoids counting a temperature allowance again when it is already included in the base statement.

A temperature allowance remains an accuracy or uncertainty term unless an authored correction model defines a signed adjustment. It does not silently change the DUT acceptance limit, and the resolver does not extend the coefficient beyond its declared conditions.

Elapsed time and instrument state distinguish a modifier from a requirement.

Conditional specifications can depend on time since calibration, warm-up duration, autocalibration status, resolution or digits, operating mode, frequency, loading and environmental conditions. A condition may select a different base statement, contribute a numerical modifier or define when a specification is valid.

Standard preserves that distinction. A missing warm-up requirement is not converted into an arbitrary extra uncertainty, and an operating-mode restriction is not treated as a coefficient. The reviewer can inspect the conditions that selected the rule, the modifiers applied and any condition that prevents the specification from being used.

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Conditional specifications and instance corrections / continued

Environmental observations and instance corrections retain their provenance.

Environmental values can apply to the run or to an individual point. Each observation retains its source, capture time and units, allowing a changing condition to be related to the point measured at that time. Missing, stale or unsuitable values remain identifiable rather than being replaced by one assumed ambient value for the entire job.

An individual instrument's calibration coefficients or certificate-derived corrections retain their own source, validity period, uncertainty, operating conditions and revision. The measurement model states how the correction applies to the result and how its uncertainty contributes. Manufacturer performance limits stay separately identifiable, so an instance correction cannot silently replace a limit or become a second copy of an already included contribution.

Deterministic rule selection and dependency tracking protect the result.

Piecewise specifications identify their ranges and the inclusion of each boundary. Selection rules make overlaps, gaps and precedence explicit, and the same instrument, point and conditions resolve to the same rule. A gap or an incompatible condition receives a named outcome instead of a guessed coefficient or hidden extrapolation.

A change to readings, conditions, specification revisions, corrections or calculation settings invalidates the draft results that depend on it and recomputes their analysis. The reviewer can inspect the inputs and rule set used for the new result before release.

Released records preserve their evaluated evidence. A later correction proceeds through controlled amendment with predecessor and successor lineage, keeping the earlier result and the reason for the change available. The laboratory can improve a model without rewriting the history of work already released.

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/specifications

MEASUREMENT ANALYSIS AND STUDY PLANNING

Understand the Calculation Before You Rely on the Result.

Explore the calculator and statistical reference tools, with the distinction between supported calculations and planned measurement-system studies made explicit.

Pilot / Early Access. Calculator and reference tools are available for demonstration. Complete MSA study workflows are planned.

The calculator library includes CMC, mathematical and statistical tools. The selected tool's inputs, assumptions and persistence determine what its result can support.

Full Gage R&R, bias, linearity, stability and attribute-agreement study workflows are planned. Their reference material does not establish a complete study workspace, report or customer-available analysis path.

Workflow and evidence

CMC Budgets	Inspect contributor inputs, coverage and the resulting capability calculation.
Statistical Tools	Confirm the specific calculator, input format and saved-result behavior in your demonstration.
Study Planning	Use the references to define the study and its acceptance criteria. Full MSA workflows remain planned.
Separate Quantities	A mean, sample standard deviation and uncertainty answer different questions. Check the supported output before applying it.

Calculators and Retained Records

Calculator drafts and saved results may live in the browser. A calculator result is distinct from the controlled laboratory record and its completed uncertainty evidence.

MEASUREMENT ANALYSIS AND STUDY PLANNING

Measurement Analysis and Study Planning / continued

A Study Needs More Than a Formula

Parts, operators, trials, methods, observations and reporting must form one accepted workflow before a complete MSA capability can be offered. Discuss the particular study you need as a future requirement; do not assume it is included in today's pilot.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST/SEMATECH Engineering Statistics Handbook, Section 2.4: Gauge R&R Studies
Study design, data collection, repeatability, reproducibility, stability and bias investigations. Chapter 2 publication 2003, online section accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/measurement-analysis

CALCULATION AND PROCESS REFERENCE

Every calculation has a purpose, a model and a reference.

Inspect the intended use, inputs, units, assumptions, formula, numerical example and limits of a calculation before applying its result to the laboratory's work.

Pilot / Early Access. Method guidance includes topics beyond current calculation and completion support. The uncertainty and measurement-analysis pages state the current boundaries.

Standard places a calculation and process reference beside its measurement tools. Each entry identifies what the calculation is for, the quantity it reports, its required inputs and conditions, and the mathematical operation it performs. A worked numerical chain lets the reader follow the inputs through intermediate values to the reported output.

The reference library covers supported calculations and planned methods. Each reference is explanatory guidance, not evidence that every described method is implemented or available in the pilot. Match it to the actual tool and retained result shown in your evaluation.

A laboratory can review the method before configuring a test, then open the same reference while inspecting a result or investigating a changed number. Formula, units and validity remain visible together, reducing the risk that a familiar statistic is applied to a different reported quantity or beyond the conditions the method supports.

Workflow and evidence

Purpose identifies the question the tool answers.	Read the intended use and the reported quantity, including the distinction between a mean, a sample standard deviation, mathematical RSS and measurement uncertainty.
Inputs retain their meaning and origin.	Inspect the required quantities, units, raw observations, source documents and evidence before accepting the calculation.
Assumptions define the valid application.	Review independence or covariance, statistical conditions, required sample structure and the operating range for which the selected method applies.

CALCULATION AND PROCESS REFERENCE

Calculation and process reference / continued

Formulas expose the calculation chain.	Follow the declared equation, sensitivity, combination, coverage and reporting operations with their intermediate values.
Numerical cases make the method reviewable.	Compare a worked input-to-output chain with the selected calculation version and its declared precision and rounding boundaries.
Authoritative references retain their edition.	Open the relevant NIST, JCGM or calibration-policy reference and identify the source version used by the method entry.
Guidance and result evidence keep separate roles.	The reference explains the calculation. The result retains the model revision, inputs and evidence that were applied to the work.
Limits prevent silent changes of meaning.	Check the method's conditions and named refusals before using a result as an uncertainty, capability claim, study conclusion or conformity decision.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

NIST/SEMATECH Engineering Statistics Handbook, Section 2.4: Gauge R&R Studies
Study design, data collection, repeatability, reproducibility, stability and bias investigations. Chapter 2 publication 2003, online section accessed 2026-10-21

CALCULATION AND PROCESS REFERENCE

Calculation and process reference / continued

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/calculation-reference

QUALITY AND RECORD CONTROL

A suspect standard can be traced to the calibrations that used it.

Quality belongs to the complete service. Connect findings, technical evidence, corrective actions, controlled documents and customer consequences.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

Standard records actual reference-standard attribution, which supports both forward and reverse traceability. A reviewer can move from a calibration to the standards supporting it, or start with a suspect standard and identify calibrations and released paper that may need investigation. The resulting set is evidence for technical judgment, not an automatic declaration that every result is invalid.

A nonconformance can retain the finding, containment, cause analysis, evidence, corrective actions and effectiveness review. Affected instruments, certificates or work orders are linked as real records. The laboratory records a decision and measurement reasoning for each affected item, with the release state of the related paper preserved.

Configured close gates check unresolved actions, undecided affected items and customer notices owed. Where a certificate must be withdrawn, the quality workflow uses the certificate process rather than changing an unrelated status field. This gives the quality manager a documented chain from the finding through the technical decision, customer consequence and closure evidence.

Workflow and evidence

Nonconformances and corrective actions	Retain the finding, containment, cause, correction and effectiveness review. Make required actions and notices part of the close criteria.
Out-of-tolerance investigations	Follow actual reference-standard use into affected calibrations and released certificates. Assess each affected result with technical judgment.
Controlled procedures and documents	Review revisions, resolve notes and retain approved copies and effective dates. Keep the applicable method with the work.

QUALITY AND RECORD CONTROL

Quality and record control / continued

Certificate review and correction	Separate technical completion, signing, QA review, generation and customer release. Link amendments or withdrawal to the earlier record and reason.
Qualifications and accountability	Relate responsibilities and permissions to the people doing and approving work. Retain the evidence needed to retrieve and defend the service.
Actual-standard reverse traceability	Find the measurements and released paper supported by a suspect standard using actual attribution and the relevant period.
A reason for each affected-item decision	Retain the technical assessment, responsible person and chosen action rather than treating every related calibration as automatically invalid.
Controlled correction and communication	Connect amendments and withdrawals with their predecessor records, customer communication and the configured delivery outcome.
Qualified responsibility	Connect signer and reviewer authority with the relevant training evidence, procedure version, expiry and retraining requirements.

Trace a suspect standard to the actual work it supported.

When a reference standard is found out of tolerance or its evidence becomes suspect, reverse traceability begins with the standards actually attributed to the measurement. The investigation identifies the affected calibration events, instruments, work orders and released certificates using that standard and the relevant period.

Quality can inspect the original standard evidence and the measurement context before deciding the consequence for each affected item. The list establishes what needs assessment; the recorded technical reasoning explains whether a result remains acceptable, requires further investigation or needs corrective action.

QUALITY AND RECORD CONTROL

Quality and record control / continued

Keep the decision, corrected paper and customer communication connected.

Each affected-item decision retains its reason and the person responsible. An amendment, correction or withdrawal follows the controlled record and certificate lifecycle, keeping the original issued document and its successor relationship visible.

The customer communication follows the laboratory's configured delivery route. The account record connects the investigation, the relevant certificate and the message or portal availability associated with the outcome. Delivery status remains a distinct fact, so staff can identify a failed notification and complete the required follow-through.

Separate performing, signing, reviewing and releasing the work.

Technical signing, independent review and release have distinct responsibilities. The person who performed the work, the person authorized to sign it and the reviewer are recorded against the relevant actions. Permissions and the laboratory's separation rules determine which role may carry each responsibility.

The review concerns the actual result and document revision being approved. A material change to the calculation, content or rendering basis returns the affected approval for review. The laboratory can retrieve who approved the version that was issued and why a later version followed it.

Connect authorization to training and the procedure revision.

Qualifications connect a person with the methods and responsibilities they are authorized to perform. Training evidence, the applicable procedure version, expiry and renewal requirements keep that authorization specific to the work rather than a general job title.

A procedure revision can introduce a retraining or requalification requirement. The work and approval paths check the applicable qualification at the point it is needed, and the responsible manager can see what must be renewed before the person resumes that responsibility.

Related website guide: metraslate.com/platform/quality

WORK ORDERS AND CONNECTED OPERATIONS

The order connects the work while custody follows the instrument.

Coordinate instrument-level service, collection, receipt, laboratory or onsite work, technical review and return through the same customer and equipment records.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

A work order organizes the agreed service into instrument-level lines. Several instruments on the same customer order can progress independently, with their own service location, technical work and next operating milestone. Staff can explain why one instrument is held while the others continue.

Physical possession and technical progress answer different questions. The equipment can be received without its calibration being finished, or its calibration can be complete while it waits for return. Connected custody, logistics and service records let the team see both parts of the job.

Workflow and evidence

Each service line identifies the work.	The instrument, service, customer, site and work-order relationship keep several pieces of equipment on one order from becoming one undifferentiated status.
The service location names a real place.	The order and its lines identify laboratory work or a site belonging to the customer, with inheritance where a line follows its order.
Collection and delivery have operational records.	Pickup and delivery stops retain their address, requested or scheduled day, customer window, notes and related work.
Custody preserves physical context.	Receipt and movement records describe possession and location separately from the technical result.
Field preparation selects the work to carry.	Selected work-order lines can be packaged for a named device with their pinned technical context and a time-limited pass.

WORK ORDERS AND CONNECTED OPERATIONS

Work orders and connected operations / continued

Review and return continue the same service history.

The calibration record, certificate and equipment return remain connected to the customer job that authorized the work.

One order can contain several different operating states.

The instrument-level line is the useful unit of service progress. The team can distinguish equipment awaiting receipt, work under way and equipment waiting for the next decision without treating the order header as a substitute for every line's state.

The work record gives customer service, laboratory staff and operations a common reference. A representative can inspect the same job the technician is working from and explain which instrument needs attention and what happens next.

Collection and return are planned against the actual customer site.

A customer can request a pickup at an authorized site with a preferred day, collection window and notes. That request creates the lab's actual logistics stop, so dispatch can continue from the submitted record.

Pickup and delivery stops can be combined into planned routes with a driver, origin, date and ordered sequence. The route and company-routes views help staff follow the next destination and recorded progress, while the instrument's physical custody remains distinct from its calibration status.

Laboratory and onsite work keep the same equipment identity.

Service location belongs to the order and its lines. Choosing a customer site connects the work to a place already recorded for that account, while a line can inherit the order's location when that is appropriate.

For disconnected field work, selected lines travel in a package containing their equipment and technical context. Captured observations return to Standard for reconciliation and uncertainty evaluation before becoming part of the completed laboratory record.

WORK ORDERS AND CONNECTED OPERATIONS

Work orders and connected operations / continued

The completed result belongs to the original job.

The manual-calibration event records the observations and supporting measurement evidence for the instrument. Completion preserves the evaluation basis, and technical review and certificate release continue from that recorded work.

Operations can then coordinate the equipment's return while the customer-facing team follows the related service and account context. The job remains understandable across the bench, the dispatch desk and the customer conversation.

Related website guide: metraslate.com/platform/operations

CUSTOMER PORTAL

Customers can manage the whole service relationship from their equipment records.

Give customers a connected place to plan service, request and approve work, arrange collection, follow their equipment, retrieve released certificates and manage the account with your laboratory.

Pilot / Early Access. A laboratory customer portal is access to that service relationship. It is distinct from an independently operated Asset workspace; payment processing is not included.

The customer portal connects the service relationship to the equipment it concerns. An authorized contact can see the instruments and sites available to them, understand what is due and open the work, decisions and documents associated with each instrument. The customer and laboratory work from the same equipment identity and service context.

Customers can request a quotation, review the laboratory's response, approve the agreed service and provide purchase-order information. They can request collection, follow shared job status, answer equipment decisions and retrieve certificates after release. Conversation, feedback, store orders, invoices and payment readiness remain part of the same account.

Workflow and evidence

Equipment and history stay connected.	Review equipment identity, authorized sites, due dates, past service and the work associated with each instrument.
Customers request and approve the work.	Prepare a quote request, review the issued quotation, approve the agreed service and retain purchase-order context.
Collection starts with a structured request.	Give the laboratory the authorized collection site, preferred date, time window and instructions.

CUSTOMER PORTAL

Customer portal / continued

The next decision stays with the instrument.	Follow the service status the laboratory shares and answer the permitted decisions for the affected equipment.
Released documents remain easy to find.	Open released certificates from the equipment record and retrieve the quotation or invoice associated with the service.
The account explains the financial position.	Review the invoices and balances your laboratory makes available. Payment processing and self-payment are not included in the validated scope described here.

Customers can plan service from their equipment and history.

The equipment record brings together the instrument's identity, site, due context, previous service and current work. Customers can identify what needs attention and begin the next service request from the instruments already on their account. The laboratory receives equipment context with the request.

Access follows the customer relationship and the contact's authorized sites. The laboratory can give a contact a useful view of their own equipment, work and documents while retaining control over information belonging to another site or account.

A quote request becomes a clear agreement about the work.

Customers select equipment, describe the required service and add the information the laboratory needs to price it. They can save and revise an unsent request, then send it when the instrument list and service needs are ready.

When the laboratory issues its quotation, the authorized contact reviews the offered service and the issued document. Approval and PO information remain with the quotation and account. A revised quotation retains its own commercial context, so the service agreement has a clear basis.

CUSTOMER PORTAL

Customer portal / continued

Pickup requests reach the laboratory's logistics workflow.

Customers request collection at an authorized site and provide a preferred day, collection window and instructions. The request becomes the pickup record the laboratory manages, with the site's address and customer context already connected.

Customers can follow the pickup information shared with them as the laboratory confirms and plans the collection. The requested date remains distinguishable from the laboratory's confirmed operational arrangement.

Customers can follow work and answer the decisions that need them.

The portal shows the service status shared for an instrument and the work order it belongs to. Customers can inspect the related visit, service progress and equipment history, then discuss that specific work with the laboratory.

When failed equipment needs a decision, the laboratory presents its permitted next steps. The authorized contact responds against the affected decision, and the answer stays with the service record. A priced repair or additional service follows its required commercial approval before the additional amount becomes collectible.

Released certificates stay with the equipment they describe.

After the laboratory releases a certificate, the customer can open it from the related equipment and service context. The equipment history provides a route back to earlier released evidence when a customer prepares for an audit, reviews a result or plans the next calibration.

Issued quotations and invoices retain their document context as well. The customer can retrieve the relevant paper while ownership, site access and release controls govern what is available.

Conversation and account services stay close to the work.

Customers can converse with the laboratory about equipment, service and orders, and provide feedback about their experience. The conversation reaches the laboratory's customer-service workflow, where staff can take responsibility and involve the colleague needed for the next answer.

CUSTOMER PORTAL

Customer portal / continued

The account also provides issued invoices, balances and purchase-order context. Where the laboratory offers customer store listings, customers can browse the permitted goods, maintain a cart, choose supported shipping or collection and follow the resulting order.

Review the Invoice and Account Position.

Authorized contacts can review invoices and balances within their customer and site permissions. A quotation approval or purchase order records an agreement; it is not a payment.

Agree the payment process with your laboratory. Card collection, temporary holds, refunds and customer self-payment require additional provider and workflow acceptance and are not offered as active capabilities here.

Follow the job to the instrument that needs your answer.

Open the work order to see how many instruments have arrived, which are on the bench, what is waiting for a decision and which documents have been released. Open an individual instrument to keep its service history, due date and certificate beside the current visit.

Related website guide: metraslate.com/customer-portal

ONSITE AND OFFLINE CALIBRATION

Carry the selected work into the field and reconcile the observations when it returns.

Prepare a device with the right work-order lines, capture raw observations locally and bring them back into Standard's uncertainty, review and certificate workflow.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

Onsite work starts with the same customer, equipment and service-line identities used by the laboratory. The order names the actual customer site, and the technician takes selected open lines onto a named device with the technical context needed to capture the work.

The field worksheet records observations locally when connectivity is unavailable. Standard calculates uncertainty, TUR and the final decision when the raw values return to the lab, then continues through the same technical review and certificate process.

Workflow and evidence

The work package identifies what left the lab.	The package retains the order, equipment, measurement points, units, limits and pinned procedure, datasheet, service-level and specification context.
The device receives a defined work pass.	A checkout identifies the holder, device, selected lines and expiry. Readiness and competing holders are checked before the package is prepared.
The field worksheet keeps raw observations locally.	Readings, attributed standards, conditions and notes enter the device's queue, preserving the information the lab will evaluate.
The lab evaluates the returned measurements.	A local limits indication helps the technician while disconnected. Uncertainty, TUR and the governing result are worked out by Standard at the lab.

ONSITE AND OFFLINE CALIBRATION

Onsite and offline calibration / continued

Reconciliation distinguishes accepted and unresolved work.	Each item retains an identity and a current lab answer, allowing retries, conflicts, refusals and completion to be handled explicitly.
Unconfirmed work can be preserved.	Save A Copy retains work the lab has not confirmed. Hand Back returns the pass while keeping unresolved evidence available for follow-up.
A controlled pass from checkout to handback	Keep the selected lines, holder, device, package, expiry, recall and takeover context connected throughout field work.
Reconciliation by observation identity	Distinguish accepted, repeated, unsent, refused and conflicting observations before completion continues.
The same technical review after the field	Evaluate accepted field evidence in Standard and continue through the normal signing, review, certificate and release lifecycle.

Preparation makes the disconnected work specific.

The service location is the laboratory or an actual site belonging to the customer. The order can supply that location to its lines, or an individual line can name its own valid service location. The work therefore remains connected to the customer's recorded place of service.

Take Offline selects open work-order lines and checks their readiness and current holder. It prepares one package for the checkout, retaining the pinned technical revisions, measurement grid, limits and standards that remain in calibration through the pass end date. The package's hash is checked when the device receives it.

The worksheet keeps what can be known in the field.

The device must be ready to keep both the application and the work locally before checkout. The technician can then record raw readings, identify the standards used, capture environmental conditions and retain notes against the packaged points.

ONSITE AND OFFLINE CALIBRATION

Onsite and offline calibration / continued

The worksheet can indicate a reading outside the limits carried with the package. It leaves uncertainty, TUR and the final conformity evaluation to Standard at the lab, where the accepted observations are evaluated against their retained technical basis.

Reconnection asks what the lab actually accepted.

While the application is open, it sends queued work on opening, reconnection or sign-in, and the technician can also choose Send Now. Each capture has its own identity. Resending an item that already landed does not create another observation, and a current reconciliation read reports what became of each item.

The receiving workflow checks the pass, selected line, package and pinned revision. Accepted readings retain their captured timing and synchronization provenance. Conflicting or refused values stay visible for attention, while accepted values remain retained even if a later item cannot be applied.

Completion follows the observations that support it.

A Finish intent is handled after the readings and supporting context. If values in that send were refused, completion waits so the technician can resolve, remove or appropriately exclude the unresolved work before the record is finished.

Once Standard completes the result, the technician can open the finished record. The laboratory can inspect the uncertainty and decision evidence, carry out its technical review and continue to certificate release from the reconciled observations.

Copies and handback keep recovery visible.

This Device distinguishes work not yet sent, values accepted by the lab and values needing attention. Save A Copy preserves unconfirmed work for recovery, while Hand Back checks the lab's current answer before ending the pass.

Work that the lab has not confirmed remains locally retained for 30 days after handback, and the copy provides another way to preserve it. Recall, takeover and a line already finished at the lab are shown as conditions to resolve, rather than a successful send being assumed from connectivity alone.

ONSITE AND OFFLINE CALIBRATION

Onsite and offline calibration / continued

Manage the whole field pass, including the work that needs recovery.

The field pass begins with selected work, an identified holder and device, a verified package and pinned technical definitions. Its expiry, recall and takeover conditions remain part of the same lifecycle as the observations captured in the field.

Each observation keeps its identity through local capture, transmission and the laboratory's reconciliation. An acknowledgment identifies what was actually accepted. Unsent, refused and conflicting work remains distinguishable, while copies and handback preserve the evidence that still needs attention.

Return accepted field evidence to the normal review path.

Completion follows the accepted observations, standards and required context. Standard evaluates the reconciled inputs and retains the uncertainty and decision evidence with the completed record. The technician and reviewer can then follow the same signing, review, certificate and release responsibilities used for laboratory work.

Reconnection does not erase the difference between a successful send and completed work. The device and laboratory retain their current standing so the technician can resolve an expired pass, a recalled line, a conflicting value or an interrupted transmission without replacing evidence already accepted.

Related website guide: metraslate.com/metrology/onsite

BILLING AND PAYMENTS

Keep the Approved Work Connected to Its Invoice.

Retain the customer's agreement, connect the invoice to the accepted work, and follow recorded receipts and balances. Payment processing has a separate adoption boundary.

Pilot / Early Access. Invoice and receipt records can be demonstrated. Active payment processing and customer self-payment are not included; confirm the accepted-agreement controls for your pilot.

Quotations, customer decisions, invoices and recorded receipts provide the commercial record around a service. A purchase order or accepted quotation records an agreement; it does not mean money has been received.

Newer agreement controls retain the reviewed estimate revision, approved subset and terms. Their availability must be confirmed against the pilot configuration. Live collection, card holds, automatic charging, refunds and customer self-payment are not active in the offer described here.

Workflow and evidence

Quotation and Approval	Keep the agreed revision and customer decision attributable to the work.
Invoice and Balance	Connect billed work, recorded receipts and the amount remaining. A recorded receipt is distinct from a processor transaction.
Customer Access	Authorized customers can view the invoices and account information made available by their laboratory.
Processing Boundary	Agree the external payment process during pilot setup. Do not assume card processing or self-payment is included.

An Approval Has a Specific Basis

The quotation gives the customer a defined service and price to review. Revised work needs its own clear decision. Retaining the agreement helps staff explain what was approved when preparing the invoice.

BILLING AND PAYMENTS

Billing and payments / continued

The Invoice Explains the Account Position

Invoice lines, recorded receipts and balances belong with the customer and work they concern. Review access and the accounting process during setup so people know which system records each fact.

Payment Automation Is a Separate Development

Configurable funding requirements, card authorizations, provider collection, refunds and self-payment need additional activation and acceptance. They are not implied by the presence of an invoice, approval or payment record. Bring these requirements to the pilot discussion.

Related website guide: metraslate.com/platform/payments

TECHNICAL RECORD

Follow the result back to its basis.

A useful evaluation inspects what was recorded, what was applied and what remains retrievable after review.

Identity	Instrument, customer and site, work-order line, controlled method and datasheet revision.
Observation	Value and unit, stimulus or nominal, phase, adjustment, environment, attributed standards and capture context.
Evaluation	Specification identity, supported contributions, coverage, TUR, guardband and decision outcome. Distinguish full calculation values from reported values.
Release and correction	Exercise review, certificate release and authorized customer access. Confirm how correction lineage, retained evidence and earlier documents behave in the selected configuration.

Current scope

Repeated-observation completion, general covariance propagation and broader scope/accreditation treatment require the separate limits described in this guide.

CAPABILITY BOUNDARIES

Know what needs its own acceptance.

The pilot is the place to inspect both supported behavior and the requirements still in development or validation.

Built-in CMC and scope budgets	CMC budgets and technical curve publication are part of the Metrology pilot discussion. Broader scope lifecycle and accreditation treatment remain in validation.
Built-in uncertainty evaluation	Supported point uncertainty is demonstrated in Standard Metrology. Repeated-observation completion and general correlated-input propagation remain outside the validated scope described here.
Measurement Analysis and Study Planning	Calculator and reference tools are available for demonstration. Complete MSA study workflows are planned.
Billing and payments	Invoice and receipt records can be demonstrated. Active payment processing and customer self-payment are not included; confirm the accepted-agreement controls for your pilot.
Integrations and technical architecture	Provider-dependent and subject to configuration and acceptance. No partner endorsement, universal connector or ready QuickBooks integration is claimed.
Analytics and Reporting	Reporting foundations can be demonstrated. Subjects, permissions and package exposure must be confirmed; a complete service-costing workflow is not claimed.

BUYING AND IMPLEMENTATION

Agree what the pilot must prove.

Bring an equipment list, representative records and a real workflow. Use the pilot to settle fit, responsibilities and the adoption decision.

Demonstrate the fit	Identify the product, edition, people, sites and workflows to evaluate. Name the customer acceptance owner and MetraSlate contact.
Agree commercial scope	Confirm enabled capabilities, licensing basis, quoted charges and inclusions. Identify separately charged implementation, import assistance, training or support. No standard price or rollout timeline is published here.
Validate a sample	Supply an authorized representative export. Current CSV and XLSX import creates new equipment records; full historical migration, relationships and attachments require separate review and acceptance.
Accept the workflows	Exercise the actual roles, permissions, record outcomes and failure cases in the pilot configuration. Keep unresolved requirements visible before wider adoption.
Complete the handover	Agree hosting and infrastructure ownership, updates, support contacts, backup operator and restore owner. Set retention and recovery expectations, then define export formats, limits, assistance and exit responsibilities.

Buying and implementation: metraslate.com/hosting

A PRACTICAL EVALUATION

Bring the work you need to prove.

Request a demonstration built around the records, decisions and people that matter to your organization.

Equipment owners	Follow identity, location, service history and supporting files. Confirm whether focused AssetLight or the broader Asset business scope fits your needs.
Calibration laboratories	Follow a customer request into the work order, recorded observation, supported calculation, review and released certificate. Inspect the retained basis, not only the final number.
Technical and operating owners	Agree configuration, permissions, integrations, migration, support, recovery and export responsibilities. A provider name or recorded receipt does not prove a live connection or processed payment.

Request a demonstration at metraslate.com/demo